# Chapter 1

# Example problem: A rigid cylinder rotating in a viscous fluid beneath a free surface.

Detailed documentation to be written. Here's the driver code...

(This problem is solved using spatially adaptive elements with a pseudo-elastic remesh strategy)

```
//LIC// ====================================================================
//LIC// This file forms part of oomph-lib, the object-oriented,
//LIC// multi-physics finite-element library, available
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//LIC//
//LIC//====================================================================
//A driver program to solve the problem of a cylinder rotating near a free
//surface

#include "generic.h"
#include "navier_stokes.h"
#include "solid.h"
#include "fluid_interface.h"

using namespace oomph;


//=start_of_namespace=================================================
/// Namespace for physical parameters
//====================================================================
namespace Global_Physical_Variables
{

 /// Pseudo-solid Poisson ratio
 double Nu=0.1;

 /// Direction of the wall normal vector
 Vector<double> Wall_normal;

 /// Function that specifies the wall unit normal
 void wall_unit_normal_fct(const Vector<double> &x,
                           Vector<double> &normal)
 {
  normal=Wall_normal;
```

```
 }

} // end_of_namespace


//My own Ellipse class
class GeneralEllipse : public GeomObject
{
private:
 //Internal data to store the centre and semi-axes
 double *centre_x_pt, *centre_y_pt, *a_pt, *b_pt;

public:
 //Constructor
 GeneralEllipse(const double &centre_x, const double &centre_y,
                const double &a, const double &b)
  : GeomObject(1,2), centre_x_pt(0), centre_y_pt(0), a_pt(0), b_pt(0)
  {
   centre_x_pt = new double(centre_x);
   centre_y_pt = new double(centre_y);
   a_pt = new double(a);
   b_pt = new double(b);
  }

 //Destructor
 ~GeneralEllipse()
  {
   delete centre_x_pt;
   delete centre_y_pt;
   delete a_pt;
   delete b_pt;
  }

 //Return the position
 void position(const Vector<double> &xi, Vector<double> &r) const
  {
   r[0] = *centre_x_pt + *a_pt*cos(xi[0]);
   r[1] = *centre_y_pt + *b_pt*sin(xi[0]);
  }
};


//A Domain
class CylinderAndInterfaceDomain : public Domain
{
public:
 double centre_x, centre_y;

private:
 double Lower_left[2], Lower_right[2], Lower_mid_left[2], Lower_mid_right[2];
 double Upper_left[2], Upper_right[2], Upper_mid_left[2], Upper_mid_right[2];
 double Lower_centre_left[2], Lower_centre_right[2];
 double Upper_centre_left[2], Upper_centre_right[2];


 /// Geometric object that represents the rotating cylinder
 GeomObject* Cylinder_pt;

public:

 //Constructor, pass the length and height of the domain
 CylinderAndInterfaceDomain(const double &Length, const double &Height)
  {

   centre_x = Length/2.0;
   centre_y = Height/2.0; //3.0*Height/4.0;
  //Create a new ellipse object to represent the rotating cylinder
   Cylinder_pt = new GeneralEllipse(centre_x,centre_y,0.2*Height,0.2*Height);

   //Set some basic coordinates

   Lower_left[0] = 0.0;
   Lower_left[1] = 0.0;

   Upper_left[0] = 0.0;
   Upper_left[1] = Height;

   Lower_right[0] = Length;
   Lower_right[1] = 0.0;

   Upper_right[0] = Length;
   Upper_right[1] = Height;


   //Let's just do some mid coordinates
   Lower_mid_left[0] = Length/10.0;
```

```
   Lower_mid_left[1] = 0.0;

   Upper_mid_left[0] = Length/10.0;
   Upper_mid_left[1] = Height;

   Vector<double> xi(1), f(2);
   xi[0] = -3.0*atan(1.0);
   Cylinder_pt->position(xi,f);

   Lower_centre_left[0] = f[0];
   Lower_centre_left[1] = f[1];

   xi[0] = 3.0*atan(1.0);
   Cylinder_pt->position(xi,f);

   Upper_centre_left[0] = f[0];
   Upper_centre_left[1] = f[1];


   Lower_mid_right[0] = 9.0*Length/10.0;
   Lower_mid_right[1] = 0.0;

   Upper_mid_right[0] = 9.0*Length/10.0;
   Upper_mid_right[1] = Height;


   xi[0] = -1.0*atan(1.0);
   Cylinder_pt->position(xi,f);

   Lower_centre_right[0] = f[0];
   Lower_centre_right[1] = f[1];

   xi[0] = 1.0*atan(1.0);
   Cylinder_pt->position(xi,f);

   Upper_centre_right[0] = f[0];
   Upper_centre_right[1] = f[1];

   //There are six macro elements
   Macro_element_pt.resize(6);

   // Build the macro elements
   for (unsigned i=0;i<6;i++)
    {Macro_element_pt[i]= new QMacroElement<2>(this,i);}
  }

 // Destructor: Empty; cleanup done in base class
 ~CylinderAndInterfaceDomain() {}

 //Private little interpolation problem
 void linear_interpolate(double Left[2], double Right[2],
                         const double &s, Vector<double> &f)
  {
   for(unsigned i=0;i<2;i++)
    {
     f[i] = Left[i] + (Right[i] - Left[i])*0.5*(s+1.0);
    }
  }



 // Sort out the vector representation of the i-th macro element
 void macro_element_boundary(const unsigned &time,
                             const unsigned &m,
                             const unsigned &direction,
                             const Vector<double> &s,
                             Vector<double>& f)
 {
  using namespace QuadTreeNames;

#ifdef WARN_ABOUT_SUBTLY_CHANGED_OOMPH_INTERFACES
   // Warn about time argument being moved to the front
   OomphLibWarning(
    "Order of function arguments has changed between versions 0.8 and 0.85",
    "CylinderAndInterfaceDomain::macro_element_boundary(...)",
    OOMPH_EXCEPTION_LOCATION);
#endif

  Vector<double> xi(1);

  //Switch on the macro element
  switch(m)
   {
    //Macro element 0, is the left-hand film
   case 0:

    switch(direction)
```

```
   {
   case N:
    linear_interpolate(Upper_left,Upper_mid_left,s[0],f);
    break;

   case S:
     linear_interpolate(Lower_left,Lower_mid_left,s[0],f);
    break;

   case W:
     linear_interpolate(Lower_left,Upper_left,s[0],f);
    break;

   case E:
     linear_interpolate(Lower_mid_left,Upper_mid_left,s[0],f);
    break;

   default:
    std::ostringstream error_stream;
    error_stream << "Direction is incorrect: " << direction << std::endl;

    throw OomphLibError(error_stream.str(),
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
   }

  break;

  //Macro element 1, is immediately left of the cylinder
 case 1:

  switch(direction)
   {
   case N:
     linear_interpolate(Upper_mid_left,Upper_centre_left,s[0],f);
    break;

   case S:
     linear_interpolate(Lower_mid_left,Lower_centre_left,s[0],f);
    break;

   case W:
     linear_interpolate(Lower_mid_left,Upper_mid_left,s[0],f);
    break;

   case E:
    xi[0] = 5.0*atan(1.0) - 2.0*atan(1.0)*0.5*(1.0+s[0]);
    Cylinder_pt->position(xi,f);
    break;

   default:
    std::ostringstream error_stream;
    error_stream << "Direction is incorrect: " << direction << std::endl;

    throw OomphLibError(error_stream.str(),
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
   }

  break;

 //Macro element 2, is immediately above the cylinder
 case 2:

  switch(direction)
   {
   case N:
     linear_interpolate(Upper_mid_left,Upper_mid_right,s[0],f);
    break;

   case S:
    xi[0] = 3.0*atan(1.0) - 2.0*atan(1.0)*0.5*(1.0+s[0]);
    Cylinder_pt->position(xi,f);
    break;

   case W:
     linear_interpolate(Upper_centre_left,Upper_mid_left,s[0],f);
    break;

   case E:
     linear_interpolate(Upper_centre_right,Upper_mid_right,s[0],f);
    break;

   default:
    std::ostringstream error_stream;
    error_stream << "Direction is incorrect: " << direction << std::endl;
```

```
    throw OomphLibError(error_stream.str(),
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
   }

  break;

  //Macro element 3, is immediately right of the cylinder
 case 3:

  switch(direction)
   {
   case N:
     linear_interpolate(Upper_centre_right,Upper_mid_right,s[0],f);
    break;

   case S:
     linear_interpolate(Lower_centre_right,Lower_mid_right,s[0],f);
    break;

   case W:
    xi[0] = -atan(1.0) + 2.0*atan(1.0)*0.5*(1.0+s[0]);
    Cylinder_pt->position(xi,f);
    break;

   case E:
     linear_interpolate(Lower_mid_right,Upper_mid_right,s[0],f);
    break;

   default:
    std::ostringstream error_stream;
    error_stream << "Direction is incorrect: " << direction << std::endl;

    throw OomphLibError(error_stream.str(),
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
   }

  break;

  //Macro element 4, is immediately below cylinder
 case 4:

  switch(direction)
   {
   case N:
    //linear_interpolate(Lower_centre_left,Lower_centre_right,s[0],f);
    xi[0] = -3.0*atan(1.0) + 2.0*atan(1.0)*0.5*(1.0+s[0]);
    Cylinder_pt->position(xi,f);
    break;

   case S:
     linear_interpolate(Lower_mid_left,Lower_mid_right,s[0],f);
    break;

   case W:
     linear_interpolate(Lower_mid_left,Lower_centre_left,s[0],f);
    break;

   case E:
     linear_interpolate(Lower_mid_right,Lower_centre_right,s[0],f);
    break;

   default:
    std::ostringstream error_stream;
    error_stream << "Direction is incorrect: " << direction << std::endl;

    throw OomphLibError(error_stream.str(),
                        OOMPH_CURRENT_FUNCTION,
                        OOMPH_EXCEPTION_LOCATION);
   }

  break;

  //Macro element 5, is right film
 case 5:

  switch(direction)
   {
   case N:
     linear_interpolate(Upper_mid_right,Upper_right,s[0],f);
    break;

   case S:
     linear_interpolate(Lower_mid_right,Lower_right,s[0],f);
```

```
      break;

     case W:
       linear_interpolate(Lower_mid_right,Upper_mid_right,s[0],f);
      break;

     case E:
       linear_interpolate(Lower_right,Upper_right,s[0],f);
      break;

     default:
      std::ostringstream error_stream;
      error_stream << "Direction is incorrect: " << direction << std::endl;

      throw OomphLibError(error_stream.str(),
                          OOMPH_CURRENT_FUNCTION,
                          OOMPH_EXCEPTION_LOCATION);
     }

    break;

   default:
    std::ostringstream error_stream;
      error_stream << "Wrong domain number: " << m<< std::endl;

      throw OomphLibError(error_stream.str(),
                          OOMPH_CURRENT_FUNCTION,
                          OOMPH_EXCEPTION_LOCATION);
   }

 }

};

//Now I need to actually create a Mesh
template<class ELEMENT>
class CylinderAndInterfaceMesh : public virtual SolidMesh
{
 double Height;

protected:
 //Pointer to the domain
 CylinderAndInterfaceDomain* Domain_pt;

public:

 //Access function to the domain
 CylinderAndInterfaceDomain* domain_pt() {return Domain_pt;}

 //Constructor,
 CylinderAndInterfaceMesh(const double &length, const double &height,
                          TimeStepper* time_stepper_pt) : Height(height)
  {
   //Create the domain
   Domain_pt = new CylinderAndInterfaceDomain(length,height);

   //Initialise the node counter
   unsigned node_count=0;
   //Vectors Used to get data from domains
   Vector<double> s(2), r(2);

   //Setup temporary storage for the Node
   Vector<Node *> Tmp_node_pt;

   //Now blindly loop over the macro elements and associate and finite
   //element with each
   unsigned Nmacro_element = Domain_pt->nmacro_element();
   for(unsigned e=0;e<Nmacro_element;e++)
    {
     //Create the FiniteElement and add to the Element_pt Vector
     Element_pt.push_back(new ELEMENT);

     //Read out the number of linear points in the element
     unsigned Np =
      dynamic_cast<ELEMENT*>(finite_element_pt(e))->nnode_1d();

     //Loop over nodes in the column
     for(unsigned l1=0;l1<Np;l1++)
      {
       //Loop over the nodes in the row
       for(unsigned l2=0;l2<Np;l2++)
        {
         //Allocate the memory for the node
         Tmp_node_pt.push_back(finite_element_pt(e)->
                               construct_node(l1*Np+l2,time_stepper_pt));

         //Read out the position of the node from the macro element
```

```
        s[0] = -1.0 + 2.0*(double)l2/(double)(Np-1);
        s[1] = -1.0 + 2.0*(double)l1/(double)(Np-1);
        Domain_pt->macro_element_pt(e)->macro_map(s,r);

        //Set the position of the node
        Tmp_node_pt[node_count]->x(0) = r[0];
        Tmp_node_pt[node_count]->x(1) = r[1];

        //Increment the node number
        node_count++;
       }
     }
   } //End of loop over macro elements
  //Now the elements have been created, but there will be nodes in
  //common, need to loop over the common edges and sort it, by reassigning
  //pointers and the deleting excess nodes

  //Read out the number of linear points in the element
  unsigned Np =
   dynamic_cast<ELEMENT*>(finite_element_pt(0))->nnode_1d();

  //DelaunayEdge between Elements 0 and 1
  for(unsigned n=0;n<Np;n++)
   {
    //Set the nodes in element 1 to be the same as in element 0
    finite_element_pt(1)->node_pt(Np*n)
     = finite_element_pt(0)->node_pt(n*Np+Np-1);

    //Remove the nodes in element 1 from the temporaray node list
    delete Tmp_node_pt[Np*Np + Np*n];
    Tmp_node_pt[Np*Np + Np*n] = 0;
   }

  //DelaunayEdge between Elements 1 and 2
  for(unsigned n=0;n<Np;n++)
   {
    //Set the nodes in element 2 to be the same as in element 1
    finite_element_pt(2)->node_pt(n*Np)
     = finite_element_pt(1)->node_pt((Np-1)*Np+Np-1-n);

    //Remove the nodes in element 2 from the temporaray node list
    delete Tmp_node_pt[2*Np*Np + n*Np];
    Tmp_node_pt[2*Np*Np + n*Np] = 0;
   }

  //DelaunayEdge between Elements 1 and 4
  for(unsigned n=0;n<Np;n++)
   {
    //Set the nodes in element 4 to be the same as in element 1
    finite_element_pt(4)->node_pt(n*Np)
     = finite_element_pt(1)->node_pt(n);

    //Remove the nodes in element 2 from the temporaray node list
    delete Tmp_node_pt[4*Np*Np + n*Np];
    Tmp_node_pt[4*Np*Np + n*Np] = 0;
   }

  //DelaunayEdge between Element 2 and 3
  for(unsigned n=0;n<Np;n++)
   {
    //Set the nodes in element 3 to be the same as in element 2
    finite_element_pt(3)->node_pt(Np*(Np-1)+n)
     = finite_element_pt(2)->node_pt(Np*n+Np-1);

    //Remove the nodes in element 3 from the temporaray node list
    delete Tmp_node_pt[3*Np*Np + Np*(Np-1)+n];
    Tmp_node_pt[3*Np*Np + Np*(Np-1)+n] = 0;
   }


  //DelaunayEdge between Element 4 and 3
  for(unsigned n=0;n<Np;n++)
   {
    //Set the nodes in element 3 to be the same as in element 4
    finite_element_pt(3)->node_pt(n)
     = finite_element_pt(4)->node_pt(Np*(Np-n-1)+Np-1);

    //Remove the nodes in element 3 from the temporaray node list
    delete Tmp_node_pt[3*Np*Np + n];
    Tmp_node_pt[3*Np*Np + n] = 0;
   }


  //DelaunayEdge between Element 3 and 5
  for(unsigned n=0;n<Np;n++)
   {
    //Set the nodes in element 5 to be the same as in element 3
```

```
   finite_element_pt(5)->node_pt(n*Np)
    = finite_element_pt(3)->node_pt(Np*n+Np-1);

   //Remove the nodes in element 5 from the temporaray node list
   delete Tmp_node_pt[5*Np*Np + n*Np];
   Tmp_node_pt[5*Np*Np + n*Np] = 0;
  }

 //Now set the actual true nodes
 for(unsigned n=0;n<node_count;n++)
  {
   if(Tmp_node_pt[n]!=0) {Node_pt.push_back(Tmp_node_pt[n]);}
  }


 //Finally set the nodes on the boundaries
 set_nboundary(5);

 for(unsigned n=0;n<Np;n++)
  {
   //Left hand side
   Node* temp_node_pt = finite_element_pt(0)->node_pt(n*Np);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(3,temp_node_pt);


   //Right hand side
   temp_node_pt = finite_element_pt(5)->node_pt(n*Np+Np-1);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(1,temp_node_pt);

   //LH part of lower boundary
   temp_node_pt = finite_element_pt(0)->node_pt(n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(0,temp_node_pt);

   //First part of upper boundary
   temp_node_pt = finite_element_pt(0)->node_pt(Np*(Np-1)+n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(2,temp_node_pt);

   //First part of hole boundary
   temp_node_pt = finite_element_pt(4)->node_pt(Np*(Np-1)+n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(4,temp_node_pt);
  }

 for(unsigned n=1;n<Np;n++)
  {
   //Middle of lower boundary
   Node* temp_node_pt = finite_element_pt(4)->node_pt(n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(0,temp_node_pt);

   //Middle of upper boundary
   temp_node_pt = finite_element_pt(2)->node_pt(Np*(Np-1)+n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(2,temp_node_pt);

   //Next part of hole
   temp_node_pt = finite_element_pt(3)->node_pt(n*Np);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(4,temp_node_pt);
  }

 for(unsigned n=1;n<Np;n++)
  {
   //Final part of lower boundary
   Node* temp_node_pt = finite_element_pt(5)->node_pt(n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(0,temp_node_pt);

   //Middle of upper boundary
   temp_node_pt = finite_element_pt(5)->node_pt(Np*(Np-1)+n);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(2,temp_node_pt);

   //Next part of hole
   temp_node_pt = finite_element_pt(2)->node_pt(Np-n-1);
   this->convert_to_boundary_node(temp_node_pt);
   add_boundary_node(4,temp_node_pt);
  }

 for(unsigned n=1;n<Np-1;n++)
  {
   //Final part of hole
   Node* temp_node_pt = finite_element_pt(1)->node_pt(Np*(Np-n-1)+Np-1);
```

```
     this->convert_to_boundary_node(temp_node_pt);
     add_boundary_node(4,temp_node_pt);
    }

   //Now loop over all the nodes and set their Lagrangian coordinates
   unsigned Nnode = nnode();
   for(unsigned n=0;n<Nnode;n++)
   {
    //Cast node to an elastic node
    SolidNode* temp_pt = static_cast<SolidNode*>(Node_pt[n]);
    for(unsigned i=0;i<2;i++)
     {temp_pt->xi(i) = temp_pt->x(i);}
   }
 }

};

//Now let's do the adaptive mesh
template<class ELEMENT>
class RefineableCylinderAndInterfaceMesh :
 public CylinderAndInterfaceMesh<ELEMENT>, public RefineableQuadMesh<ELEMENT>
{
public:

 // Constructor
 RefineableCylinderAndInterfaceMesh(const double &length, const double &height,
                                    TimeStepper* time_stepper_pt) :
  CylinderAndInterfaceMesh<ELEMENT>(length,height,time_stepper_pt)
  {

   // Nodal positions etc. were created in constructor for
   // Cylinder...<...>. Need to setup adaptive information.

   // Loop over all elements and set macro element pointer
   for (unsigned e=0;e<6;e++)
    {
     dynamic_cast<ELEMENT*>(this->element_pt(e))->
      set_macro_elem_pt(this->Domain_pt->macro_element_pt(e));
    }

   // Setup quadtree forest for mesh refinement
   this->setup_quadtree_forest();

   // Setup the boundary element info
   this->setup_boundary_element_info();

  }


 /// Destructor: Empty
 virtual ~RefineableCylinderAndInterfaceMesh() {}


};

template<class ELEMENT>
class RefineableRotatingCylinderProblem : public Problem
{
private:
 double Length, Height;
 //Constitutive law used to determine the mesh deformation
 ConstitutiveLaw *Constitutive_law_pt;

 Data* Traded_pressure_data_pt;

public:

 double Re, Ca, ReInvFr, Bo;

 double Omega;
 double Volume;

 double Angle;

 Vector<double> G;

 /// Constructor: Pass flag to indicate if you want
 /// a constant source function or the tanh profile.
 RefineableRotatingCylinderProblem(const double &length, const double &height);

 /// Update the problem specs after solve (empty)
 void actions_after_newton_solve() {}

 /// Update the problem specs before solve:
 void actions_before_newton_solve() {set_boundary_conditions();}

 /// Strip off the interface before adaptation
```

```
void actions_before_adapt()
 {
  this->delete_volume_constraint_elements();
  this->delete_free_surface_elements();
 }

void actions_after_adapt() {finish_problem_setup(); this->rebuild_global_mesh();}

/// Complete problem setup: Setup element-specific things
/// (source fct pointers etc.)
void finish_problem_setup();

//Access function for the mesh
RefineableCylinderAndInterfaceMesh<ELEMENT>* Bulk_mesh_pt;

//Access function for surface mesh
Mesh* Surface_mesh_pt;

//Access function for point mesh
Mesh* Point_mesh_pt;

/// The volume constraint mesh
Mesh* Volume_constraint_mesh_pt;

void set_boundary_conditions();

void solve();

/// Create the volume constraint elements
void create_volume_constraint_elements()
 {
  //The single volume constraint element
  VolumeConstraintElement* vol_constraint_element =
   new VolumeConstraintElement(&Volume,Traded_pressure_data_pt,0);
  Volume_constraint_mesh_pt->add_element_pt(vol_constraint_element);

  //Loop over all boundaries (or just 1 and 2 why?)
  for(unsigned b=0;b<4;b++)
   {
    // How many bulk fluid elements are adjacent to boundary b?
    unsigned n_element = Bulk_mesh_pt->nboundary_element(b);

    // Loop over the bulk fluid elements adjacent to boundary b?
    for(unsigned e=0;e<n_element;e++)
     {
      // Get pointer to the bulk fluid element that is
      // adjacent to boundary b
      ELEMENT* bulk_elem_pt = dynamic_cast<ELEMENT*>(
       Bulk_mesh_pt->boundary_element_pt(b,e));

      //Find the index of the face of element e along boundary b
      int face_index = Bulk_mesh_pt->face_index_at_boundary(b,e);

      // Create new element
      ElasticLineVolumeConstraintBoundingElement<ELEMENT>* el_pt =
       new ElasticLineVolumeConstraintBoundingElement<ELEMENT>(
        bulk_elem_pt,face_index);

      //Set the "master" volume control element
      el_pt->set_volume_constraint_element(vol_constraint_element);

      // Add it to the mesh
      Volume_constraint_mesh_pt->add_element_pt(el_pt);
     }
   }
 }

void delete_volume_constraint_elements()
 {
  unsigned n_element = Volume_constraint_mesh_pt->nelement();
  for(unsigned e=0;e<n_element;e++)
   {
    delete Volume_constraint_mesh_pt->element_pt(e);
   }
  Volume_constraint_mesh_pt->flush_element_and_node_storage();
 }
void create_free_surface_elements()
 {
  //Find number of elements adjacent to upper boundary
  unsigned n_boundary_element = Bulk_mesh_pt->nboundary_element(2);
  //The boundary elements do no necessarily come in order, so we will
  //need to detect the element adjacent to boundary 1.
  //The index of that element in our array will be stored in this variable
  //(initialised to a negative and therefore invalid number)
  int final_element_index=-1;
  //Loop over the elements adjacent to the boundary
  for(unsigned e=0;e<n_boundary_element;e++)
```

```
    {
     //Create the free surface element (on face 2)
     FiniteElement *free_surface_element_pt
      = new ElasticLineFluidInterfaceElement<ELEMENT>
      (Bulk_mesh_pt->boundary_element_pt(2,e),
       Bulk_mesh_pt->face_index_at_boundary(2,e));
     //Push it back onto the stack
     Surface_mesh_pt->add_element_pt(free_surface_element_pt);

      //Check whether the element is on the boundary 1
      unsigned n_node = free_surface_element_pt->nnode();
      //Only need to check the end nodes
      if((free_surface_element_pt->node_pt(0)->is_on_boundary(1)) ||
         (free_surface_element_pt->node_pt(n_node-1)->is_on_boundary(1)))
       {
          final_element_index=e;
       }
    }

   unsigned Nfree = Surface_mesh_pt->nelement();
   oomph_info << Nfree << " free surface elements assigned" << std::endl;

    if(final_element_index == -1)
     {
      throw OomphLibError("No Surface Element adjacent to boundary 1\n",
                          OOMPH_CURRENT_FUNCTION,
                          OOMPH_EXCEPTION_LOCATION);
     }

    //Make the edge point
    FiniteElement* point_element_pt=
     dynamic_cast<ElasticLineFluidInterfaceElement<ELEMENT>*>
     (Surface_mesh_pt->element_pt(final_element_index))
     ->make_bounding_element(1);

    //Add it to the stack
    Point_mesh_pt->add_element_pt(point_element_pt);
  }

 //Function to delete the free surface elements
 void delete_free_surface_elements()
  {
   //Find the number of traction elements
   unsigned Nfree_surface = Surface_mesh_pt->nelement();

   //The traction elements are ALWAYS? stored at the end
   //So delete and remove them, add one to get rid of the constraint
   for(unsigned e=0;e<Nfree_surface;e++)
    {
     delete Surface_mesh_pt->element_pt(e);
    }
   Surface_mesh_pt->flush_element_and_node_storage();

   delete Point_mesh_pt->element_pt(0);
   Point_mesh_pt->flush_element_and_node_storage();
  }


};


//============================================================================
/// Constructor for adaptive Poisson problem in deformable fish-shaped
/// domain. Pass bool to indicate if we want a constant source
/// function or the one that produces a tanh step.
//============================================================================
template<class ELEMENT>
RefineableRotatingCylinderProblem<ELEMENT>::RefineableRotatingCylinderProblem(
 const double &length, const double &height) : Length(length), Height(height),
                                               Re(0.0), Ca(0.001),
                                               ReInvFr(0.0),
                                               Bo(0.0), Omega(1.0),
                                               Volume(12.0),
                                               Angle(1.57)
{
 Global_Physical_Variables::Wall_normal.resize(2);
 Global_Physical_Variables::Wall_normal[0] = 1.0;
 Global_Physical_Variables::Wall_normal[1] = 0.0;

 G.resize(2);
 G[0] = 0.0; G[1] = -1.0;

 /// Set the initial value of the ReInvFr = Bo/Ca
 ReInvFr = Bo/Ca;

 /// Build a linear solver: Use HSL's MA42 frontal solver
```

```
 //linear_solver_pt() = new HSL_MA42;

 //Set the constituive law
 Constitutive_law_pt = new GeneralisedHookean(&Global_Physical_Variables::Nu);


 /// Switch off full doc for frontal solver
 //static_cast<HSL_MA42*>(linear_solver_pt())->disable_doc_stats();

 //Allocate the timestepper (no timedependence)
 add_time_stepper_pt(new Steady<0>);

 // Build mesh
 Bulk_mesh_pt=
  new RefineableCylinderAndInterfaceMesh<ELEMENT>(length,height,
                                                  Problem::time_stepper_pt());
 // Set error estimator
 Z2ErrorEstimator* error_estimator_pt=new Z2ErrorEstimator;
 Bulk_mesh_pt->spatial_error_estimator_pt()=error_estimator_pt;


 //Refine the problem a couple of times
 bool update_all_solid_nodes=true;
 Bulk_mesh_pt->refine_uniformly();
 Bulk_mesh_pt->node_update(update_all_solid_nodes);
 Bulk_mesh_pt->refine_uniformly();
 Bulk_mesh_pt->node_update(update_all_solid_nodes);
 //Bulk_mesh_pt->refine_uniformly();
 //refine_uniformly();
 //refine_uniformly();

 // Loop over all elements and unset macro element pointer
 unsigned Nelement = Bulk_mesh_pt->nelement();
 for(unsigned e=0;e<Nelement;e++)
  {
   dynamic_cast<ELEMENT*>(Bulk_mesh_pt->element_pt(e))->
    set_macro_elem_pt(0);
  }


 //The external pressure is a piece of global data
 Traded_pressure_data_pt = new Data(1);
 this->add_global_data(Traded_pressure_data_pt);

 // Complete the build of all elements so they are fully functional

 Surface_mesh_pt = new Mesh;
 Point_mesh_pt = new Mesh;
 Volume_constraint_mesh_pt = new Mesh;

 finish_problem_setup();

 this->add_sub_mesh(Bulk_mesh_pt);
 this->add_sub_mesh(Surface_mesh_pt);
 this->add_sub_mesh(Point_mesh_pt);
 this->add_sub_mesh(Volume_constraint_mesh_pt);

 this->build_global_mesh();

 //Attach the boundary conditions to the mesh
 oomph_info <<"Number of equations: " << assign_eqn_numbers() << std::endl;

}


//========================================================================
/// Complete build of Poisson problem:
/// Loop over elements and setup pointers to source function
///
//========================================================================
template<class ELEMENT>
void RefineableRotatingCylinderProblem<ELEMENT>::finish_problem_setup()
{
 //Now sort out the free surface
 this->create_free_surface_elements();
 //Create the volume constraint elements
 this->create_volume_constraint_elements();
 // Set the boundary conditions for this problem: All nodes are
 // free by default -- just pin the ones that have Dirichlet conditions
 // here.

 //Pin bottom and cylinder
  unsigned num_bound = Bulk_mesh_pt->nboundary();
  for(unsigned ibound=0;ibound<num_bound;ibound+=4)
   {
    unsigned num_nod= Bulk_mesh_pt->nboundary_node(ibound);
    for (unsigned inod=0;inod<num_nod;inod++)
```

```
    {
     Bulk_mesh_pt->boundary_node_pt(ibound,inod)->pin(0);
     Bulk_mesh_pt->boundary_node_pt(ibound,inod)->pin(1);
    }
  }

 //Pin u and v on LHS
 {
  unsigned num_nod= Bulk_mesh_pt->nboundary_node(3);
  for (unsigned inod=0;inod<num_nod;inod++)
   {
    Bulk_mesh_pt->boundary_node_pt(3,inod)->pin(0);
    //Bulk_mesh_pt->boundary_node_pt(3,inod)->pin(1);
   }
 }

 //Pin u and v on RHS
 {
  unsigned num_nod= Bulk_mesh_pt->nboundary_node(1);
  for (unsigned inod=0;inod<num_nod;inod++)
   {
    Bulk_mesh_pt->boundary_node_pt(1,inod)->pin(0);
    Bulk_mesh_pt->boundary_node_pt(1,inod)->pin(1);
   }
 }


 dynamic_cast<FluidInterfaceBoundingElement*>
  (Point_mesh_pt->element_pt(0))->set_contact_angle(&Angle);

 dynamic_cast<FluidInterfaceBoundingElement*>
  (Point_mesh_pt->element_pt(0))->ca_pt() = &Ca;


 dynamic_cast<FluidInterfaceBoundingElement*>
  (Point_mesh_pt->element_pt(0))->
  wall_unit_normal_fct_pt() = &Global_Physical_Variables::wall_unit_normal_fct;

 //Pin one pressure
 dynamic_cast<ELEMENT*>(Bulk_mesh_pt->element_pt(0))->fix_pressure(0,0.0);

 //Loop over the lower boundary and pin nodal positions in both directions
 unsigned num_nod= Bulk_mesh_pt->nboundary_node(0);
 for (unsigned inod=0;inod<num_nod;inod++)
  {
   Bulk_mesh_pt->boundary_node_pt(0,inod)->pin_position(0);
   Bulk_mesh_pt->boundary_node_pt(0,inod)->pin_position(1);
  }

 //Loop over the RHS side and pin in x and y
 num_nod= Bulk_mesh_pt->nboundary_node(1);
 for (unsigned inod=0;inod<num_nod;inod++)
  {
   Bulk_mesh_pt->boundary_node_pt(1,inod)->pin_position(0);
   //Bulk_mesh_pt->boundary_node_pt(1,inod)->pin_position(1);
 }


//Loop over the LHS side and pin in x
 num_nod= Bulk_mesh_pt->nboundary_node(3);
for (unsigned inod=0;inod<num_nod;inod++)
 {
  Bulk_mesh_pt->boundary_node_pt(3,inod)->pin_position(0);
  //Bulk_mesh_pt->boundary_node_pt(3,inod)->pin_position(1);
 }
//Loop over the cylinder and pin nodal positions in both directions
num_nod= Bulk_mesh_pt->nboundary_node(4);
for (unsigned inod=0;inod<num_nod;inod++)
 {
  Bulk_mesh_pt->boundary_node_pt(4,inod)->pin_position(0);
  Bulk_mesh_pt->boundary_node_pt(4,inod)->pin_position(1);
 }


//Find number of elements in mesh
unsigned Nfluid = Bulk_mesh_pt->nelement();
//Find the number of free surface elements
unsigned Nfree = Surface_mesh_pt->nelement();

   // Loop over the elements to set up element-specific
   // things that cannot be handled by constructor
   for(unsigned i=0;i<Nfluid;i++)
    {
     // Upcast from FiniteElement to the present element
     ELEMENT *temp_pt = dynamic_cast<ELEMENT*>(Bulk_mesh_pt->element_pt(i));

     //Set the source function pointer
```

```
      temp_pt->re_pt() = &Re;
      temp_pt->re_invfr_pt() = &ReInvFr;
      temp_pt->g_pt() = &G;

      //Assign the mesh deformation constitutive law
      temp_pt->constitutive_law_pt() = Constitutive_law_pt;

     }


    // Pin the redundant solid pressures (if any)
    PVDEquationsBase<2>::pin_redundant_nodal_solid_pressures(
     Bulk_mesh_pt->element_pt());

    //Loop over the free surface elements
    for(unsigned i=0;i<Nfree;i++)
     {
      // Upcast from FiniteElement to the present element
      ElasticLineFluidInterfaceElement<ELEMENT> *temp_pt =
       dynamic_cast<ElasticLineFluidInterfaceElement<ELEMENT>*>
       (Surface_mesh_pt->element_pt(i));
      //Set the Capillary number
      temp_pt->ca_pt() = &Ca;

      //Pass the Data item that contains the external pressure
      temp_pt->set_external_pressure_data(this->global_data_pt(0));
     }

}

template<class ELEMENT>
void RefineableRotatingCylinderProblem<ELEMENT>::set_boundary_conditions()
{
 //Only bother to set non-zero velocity on the cylinder
 unsigned Nnode = Bulk_mesh_pt->nboundary_node(4);
 for(unsigned n=0;n<Nnode;n++)
  {
   //Get x and y
   double x = Bulk_mesh_pt->boundary_node_pt(4,n)->x(0);
   double y = Bulk_mesh_pt->boundary_node_pt(4,n)->x(1);

   //Now find the vector distance to the centre
   double len_x = x - Bulk_mesh_pt->domain_pt()->centre_x;
   double len_y = y - Bulk_mesh_pt->domain_pt()->centre_y;

   //Calculate the angle and radius
   double r = sqrt(len_x*len_x + len_y*len_y);
   double theta = atan2(len_y,len_x);

   //Now set the velocities
   Bulk_mesh_pt->boundary_node_pt(4,n)->set_value(0,-Omega*r*sin(theta));
   Bulk_mesh_pt->boundary_node_pt(4,n)->set_value(1, Omega*r*cos(theta));
  }
}

template<class ELEMENT>
void RefineableRotatingCylinderProblem<ELEMENT>::solve()
{
 Newton_solver_tolerance = 1.0e-8;
 //Document the solution
 std::ofstream filenamee("input.dat");
 Bulk_mesh_pt->output(filenamee,5);
 Surface_mesh_pt->output(filenamee,5);
 //Point_mesh_pt->output(filenamee,5);
 filenamee.close();

 //Solve the initial value problem
 newton_solve();

 std::ofstream filename("first.dat");
 Bulk_mesh_pt->output(filename,5);
 Surface_mesh_pt->output(filename,5);
 //Point_mesh_pt->output(filename,5);
 filename.close();

 //Initialise the value of the arc-length
 double ds=0.001;
 std::ofstream trace("trace.dat");

 trace << Ca << " " << ReInvFr << " "
       << Bulk_mesh_pt->boundary_node_pt(2,0)->x(1) << std::endl;

// bool flag=true, fflag=true;

 for(unsigned i=0;i<2;i++)
  {
   if(i<5)
```

```
    {
     //Decrease the contact angle
     Angle -= 0.1;
     newton_solve(2);
     //newton_solve();
    }
   else
    {
     //do an arc-length continuation step in Ca
     ds = arc_length_step_solve(&Ca,ds);
    }

  //  if(flag)
//     {
//      //Do an arc-length continuation step in ReInvFr
//      ds = arc_length_step_solve(&ReInvFr,ds);
//     }
//    else
//     {
//      //Reset arc-length parameters
//      if(fflag) {reset_arc_length_parameters(); fflag=false;}
//      ds = 0.001;
//      //Now do it in Ca
//      ds = arc_length_step_solve(&Ca,ds);
//     }

//    if(Bulk_mesh_pt->boundary_node_pt(2,0)->x(1) < 4.0)
//     {flag=false;}

   trace « Ca « " " « ReInvFr « " " « Angle « " "
         « Bulk_mesh_pt->boundary_node_pt(2,0)->x(1) « std::endl;

   char file[100];
   snprintf(file, sizeof(file), "step%i.dat",i);
   filename.open(file);
   Bulk_mesh_pt->output(filename,5);
   Surface_mesh_pt->output(filename,5);
   //Point_mesh_pt->output(filename,5);
   filename.close();

   //Now reset the values of the lagrange multipliers and the xi's
   //An updated lagrangian approach

   //Now loop over all the nodes and set their Lagrangian coordinates
   unsigned Nnode = Bulk_mesh_pt->nnode();
   for(unsigned n=0;n<Nnode;n++)
    {
     //Cast node to an elastic node
     SolidNode* temp_pt = static_cast<SolidNode*>(Bulk_mesh_pt->node_pt(n));
     for(unsigned j=0;j<2;j++) {temp_pt->xi(j) = temp_pt->x(j);}
    }

   //Find the number of free surface elements
   unsigned Nfree = Surface_mesh_pt->nelement();
   //Loop over the free surface elements
   for(unsigned n=0;n<Nfree;n++)
    {
     // Upcast from FiniteElement to the present element
     ElasticLineFluidInterfaceElement<ELEMENT> *temp_pt =
      dynamic_cast<ElasticLineFluidInterfaceElement<ELEMENT>*>
      (Surface_mesh_pt->element_pt(n));
     unsigned Nnode = temp_pt->nnode();
     //Reset the lagrange multipliers
     for(unsigned j=0;j<Nnode;j++) {temp_pt->lagrange(j) = 0.0;}
    }
  }
 //Document the solution
 //filename.open("output.dat");
 //Bulk_mesh_pt->output(filename,5);
 //filename.close();
 trace.close();
}


////////////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////////////
////////////////////////////////////////////////////////////////////////


 int main()
  {

   RefineableRotatingCylinderProblem
    <RefineablePseudoSolidNodeUpdateElement<RefineableQCrouzeixRaviartElement<2>,
    RefineableQPVDElementWithContinuousPressure<2> > > problem(3.0,4.0);

   //ofstream filename("mesh.dat");
```

```
  //problem.Bulk_mesh_pt->output(filename,5);

  problem.solve();
 }
```

## 1.1 PDF file

A pdf version of this document is available. \